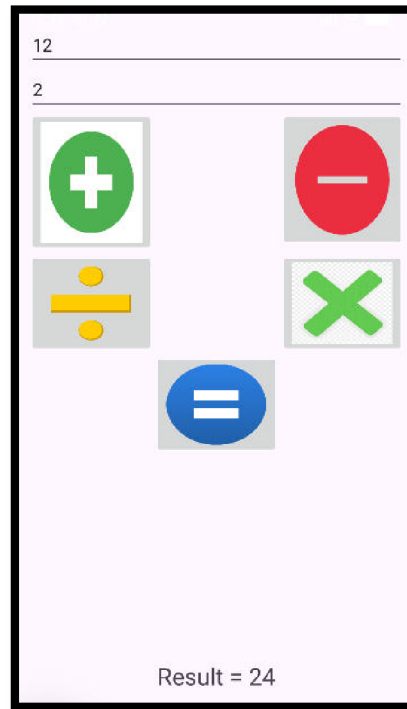


Experiment No - 12

Design an Android application to create a calculator app using GridLayout. Perform arithmetic operations such as addition, subtraction, multiplication, and division using ImageButtons.

Output –



Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<GridLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp"
    android:columnCount="2">

    <EditText
        android:id="@+id/etnumber1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter number 1"
        android:inputType="number"
        android:layout_columnSpan="2" />

    <EditText
        android:id="@+id/etnumber2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter number 2"
        android:inputType="number"
        android:layout_columnSpan="2" />
```

```
<ImageButton
    android:id="@+id/imgplus"
    android:layout_width="126dp"
    android:layout_height="142dp"
    android:scaleType="fitXY"
    android:scaleX="1"
    android:scaleY="1"
    app:srcCompat="@drawable/plus" />
```

```
<ImageButton
    android:id="@+id/imgminus"
    android:layout_width="125dp"
    android:layout_gravity="right"
    android:layout_height="137dp"
    android:scaleType="fitXY"
    android:scaleX="1"
    android:scaleY="1"
    app:srcCompat="@drawable/minus" />
```

```
<ImageButton
    android:id="@+id/imgdivide"
    android:layout_width="126dp"
    android:layout_height="102dp"
    android:contentDescription="Divide"
    android:scaleType="fitXY"
    android:scaleX="1"
    android:scaleY="1"
    app:srcCompat="@drawable/divide" />
```

```
<ImageButton
    android:id="@+id/imgmult"
    android:layout_width="125dp"
    android:layout_gravity="right"
    android:layout_height="102dp"
    android:scaleType="fitXY"
    android:scaleX="1"
    android:scaleY="1"
    app:srcCompat="@drawable/mult" />
```

```
<ImageButton
    android:id="@+id/imgequal"
    android:layout_width="126dp"
    android:layout_height="102dp"
    android:layout_columnSpan="2"
    android:layout_gravity="center"
    android:scaleType="fitXY"
    android:scaleX="1"
    android:scaleY="1"
    app:srcCompat="@drawable/equals" />
```

```
<TextView
    android:id="@+id/txtresult"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:textSize="24sp"
    android:layout_gravity="center"
```

```
        android:text="Result"
        android:layout_columnSpan="2" />
```

```
</GridLayout>
```

MainActivity.java

```
package com.example.calculator;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.EditText;
import android.widget.ImageButton;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {

    EditText e1, e2;
    ImageButton b1, b2, b3, b4, b5;
    TextView t1;
    int result = 0;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        e1 = findViewById(R.id.etnumber1);
        e2 = findViewById(R.id.etnumber2);
        b1 = findViewById(R.id.imgplus);
        b2 = findViewById(R.id.imgminus);
        b3 = findViewById(R.id.imgdivide);
        b4 = findViewById(R.id.imgmult);
        b5 = findViewById(R.id.imgequal);
        t1 = findViewById(R.id.txtresult);

        b1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                int n1 = Integer.parseInt(e1.getText().toString());
                int n2 = Integer.parseInt(e2.getText().toString());
                result = n1 + n2;
            }
        });

        b2.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                int n1 = Integer.parseInt(e1.getText().toString());
                int n2 = Integer.parseInt(e2.getText().toString());
                result = n1 - n2;
            }
        });

        b3.setOnClickListener(new View.OnClickListener() {
            @Override
```

```

        public void onClick(View v) {
            int n1 = Integer.parseInt(e1.getText().toString());
            int n2 = Integer.parseInt(e2.getText().toString());
            result = n1 / n2;
        }
    });

    b4.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            int n1 = Integer.parseInt(e1.getText().toString());
            int n2 = Integer.parseInt(e2.getText().toString());
            result = n1 * n2;
        }
    });

    b5.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            t1.setText("Result = " + result);
        }
    });
}
}

```